

- ① next element
- ② maximum sub sum
- ③ help same
- ④ calculate the sum
- ⑤ maximum sum path in array

3 4 12 8 7 6 4 3 2 1

$P \leftarrow arr[i] \leftarrow arr[i+1]$

$n=2$

$p=3$

① P ✓

② q ✓

③ P 8a

④ PH to n-1

arr

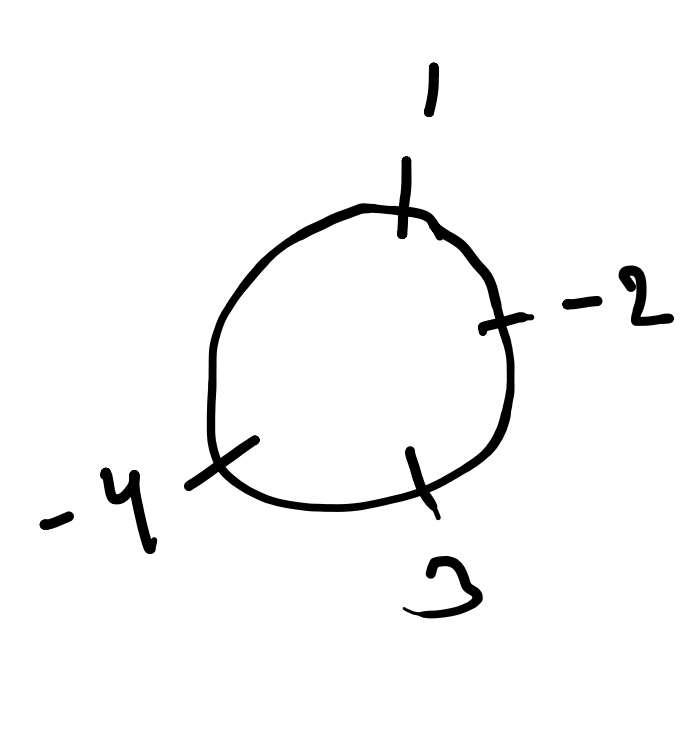
3 4 1 4 1 2 3 3 7 7 8

3 4 1 4 1 2 3 3 7 7 8

3 4 1 4 1 2 3 3 7 7 8

[1 -2 3 -4]

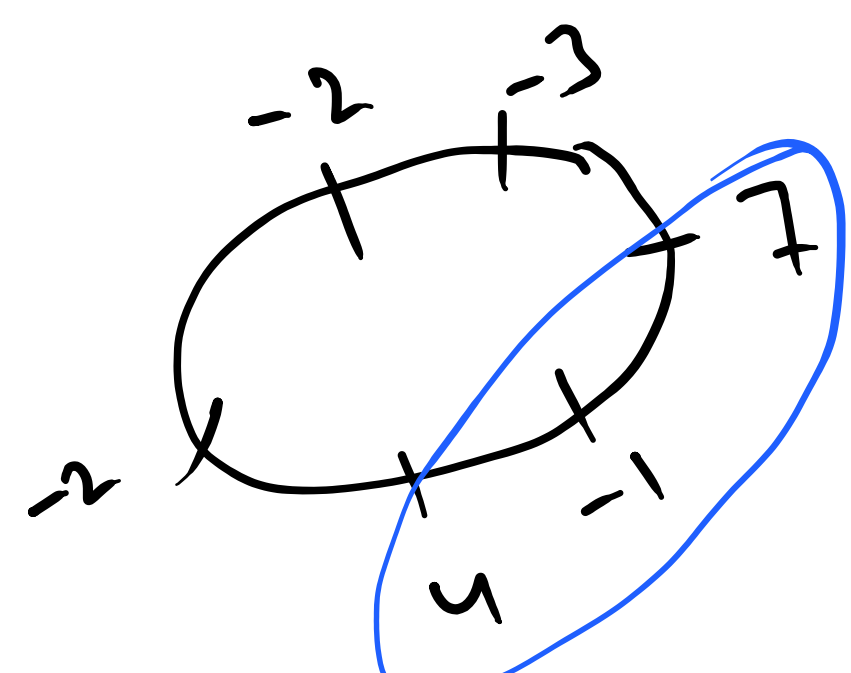
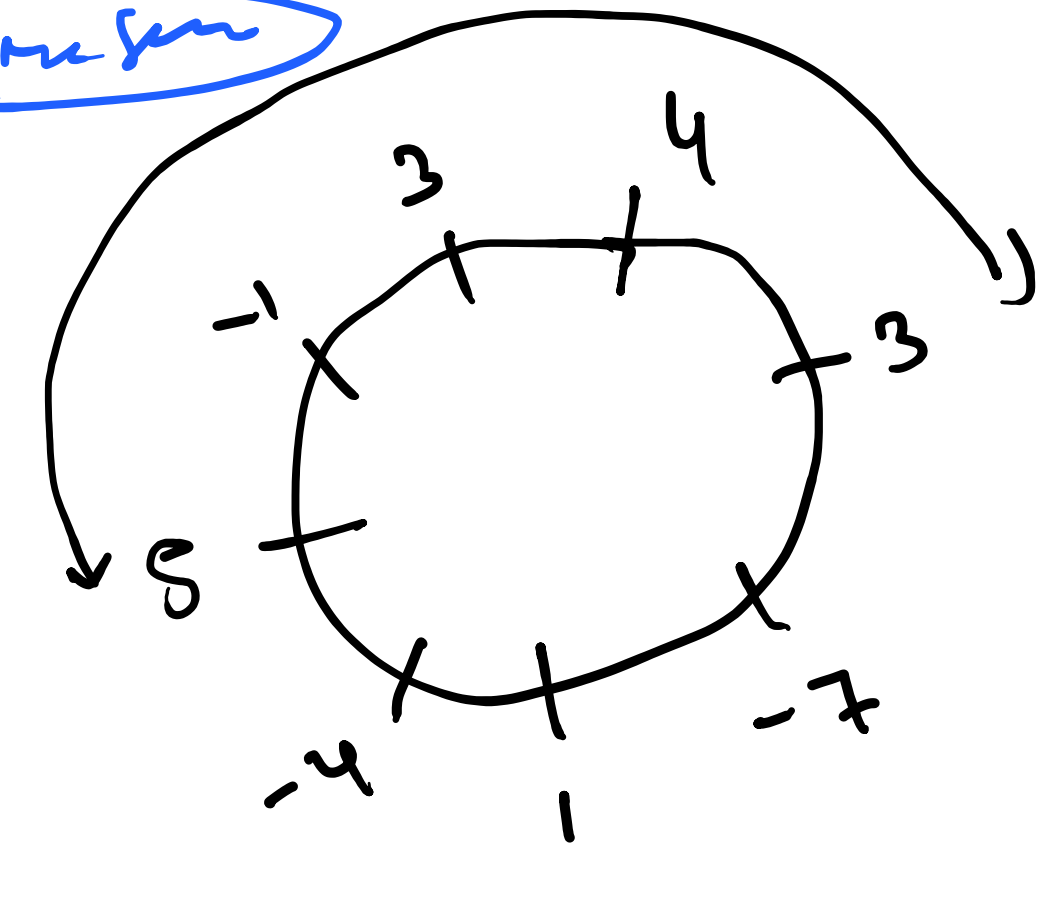
-4, 1
-4, -1, -2
-4, -1, -2



3
3-4
3-4+1
3-4+1-1

① [4 3 -7 1 -4 5 -1 3]

② [-2, -3, 7, -1, 4, -2] → Kadane

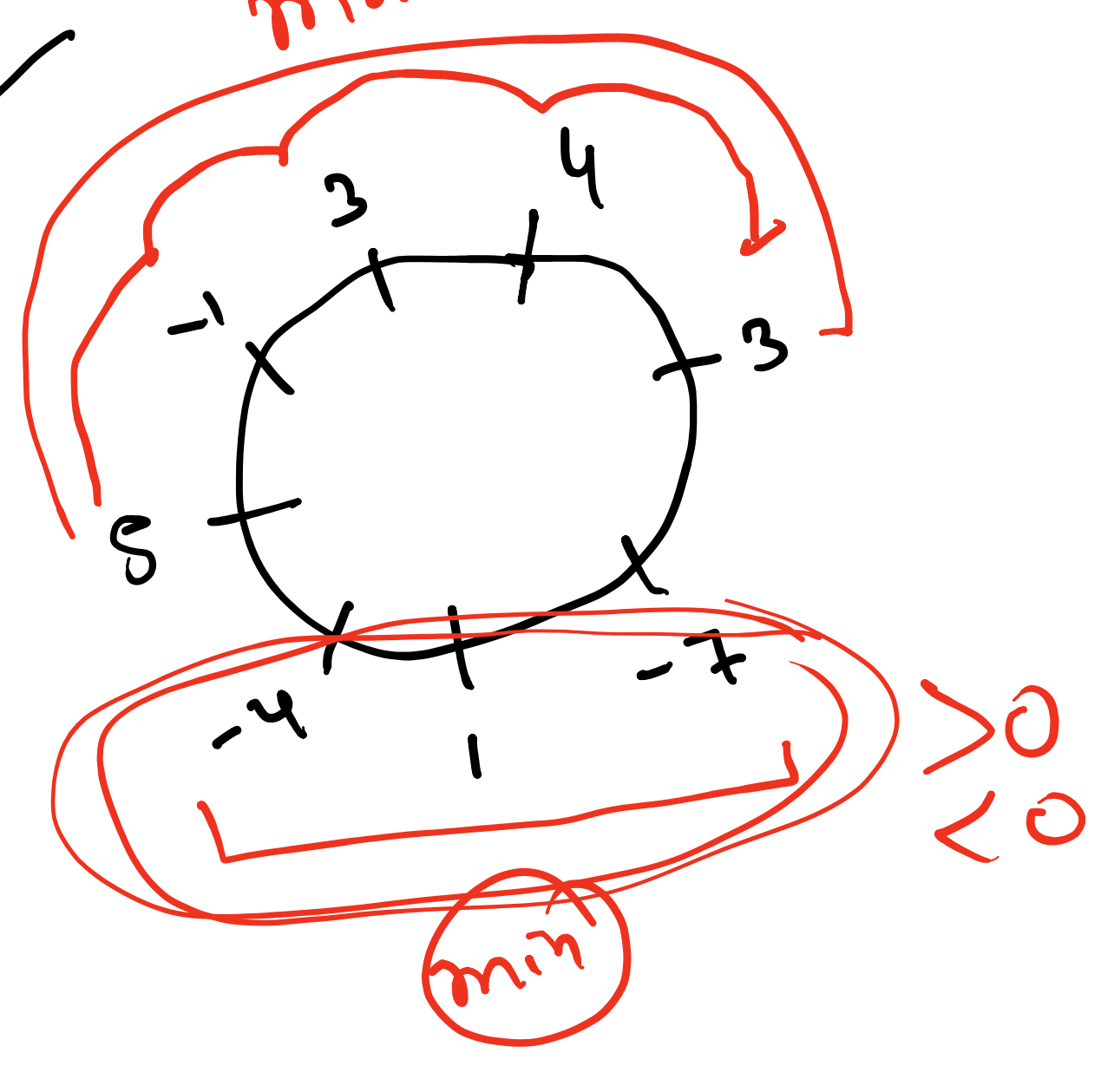


4
4 + 3 + (-7 + 1 - 4) + 5 - 1 + 3 + (-10) = 10

[4 3 -7 1 -4 5 -1 3]

[-4 -3 7 -1 4 -5 + 1 -3]

10



```
public static int Circular_Subarray(int[] nums) {  
    int linear_sum = Kadanes_Algorithm(nums);  
    int total_sum = 0;  
    for (int i = 0; i < nums.length; i++) {  
        total_sum += nums[i];  
        nums[i] = -1 * nums[i];  
    }  
    int middle_sum = Kadanes_Algorithm(nums);  
    int Circular_sum = total_sum + middle_sum;  
    return Math.max(linear_sum, Circular_sum);  
}
```

[-2 -1 -3] = -6

[2 1 3] = 6

0

① cub
1 km

② rickshaw
1 km

c1 c2 c3 c4

1 3 7 19

2 3

2 5

4 4 4

2
4
5

$2 \times c_1 = (2 \times 1, 3) = 2$

$5 \times c_4 = (5 \times 1, 3) = 3$

$(5, 7) = 8$

$(4 \times 1, 3) = 3$

$(4 \times 1, 3) = 3$

$(4 \times 1, 3) = 3$

$9 \rightarrow 1 = 7$

min [12, 14] = 12

8 8

2 3 7 10 12 15 30 34

1 5 7 8 10 15 16 19

s1

s2

1 5 7 = 13 + 18 + 27 + 64

s1-i ✓ 10 27

s2-j ✓ 18 15

```
public static int Maximum_Score(int[] arr1, int[] arr2) {  
    int s1 = 0, s2 = 0, i = 0, j = 0;  
    long sum = 0;  
    int mod = 1000_000_007;  
    while (i < arr1.length && j < arr2.length) {  
        if (arr1[i] < arr2[j]) {  
            i++;  
        } else if (arr1[i] > arr2[j]) {  
            j++;  
        } else {  
            // s1 to i  
            long sum1 = sum_of_Array(arr1, s1, i);  
            // s2 to j  
            long sum2 = sum_of_Array(arr2, s2, j);  
            sum += Math.max(sum1, sum2);  
            i++; j++;  
            s1 = i, s2 = j;  
        }  
    }  
}
```

sum = 13 + 18 + 27

2 3 7 10 12 15 30 34

1 5 7 8 10 15 16 19

s1

s2

12 3 4 5

12 3 4 5

12 3 4 5

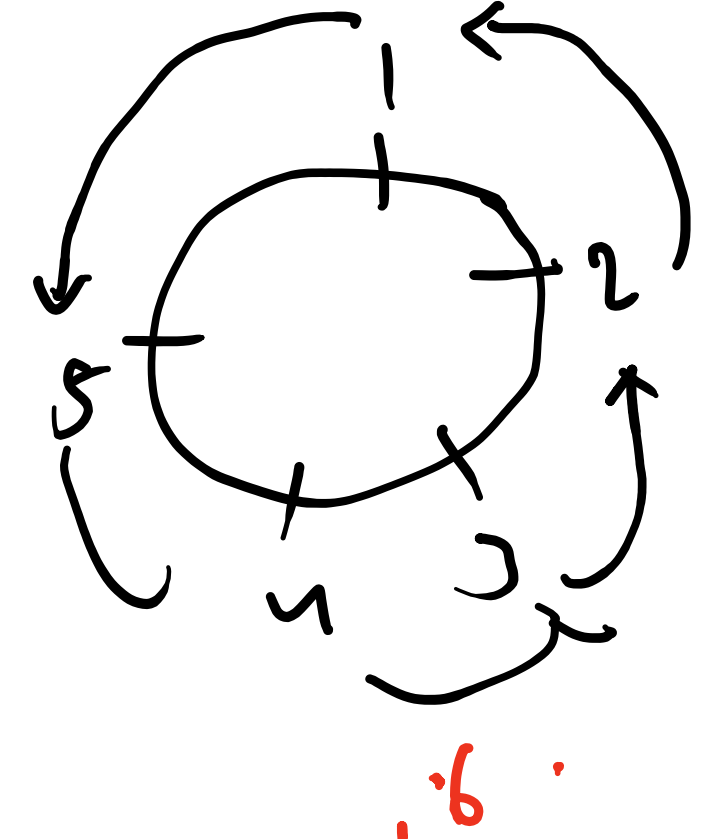
12 3 4 5

1 2 3 4 5

6 3 5 7 9

6 13 15 17 19

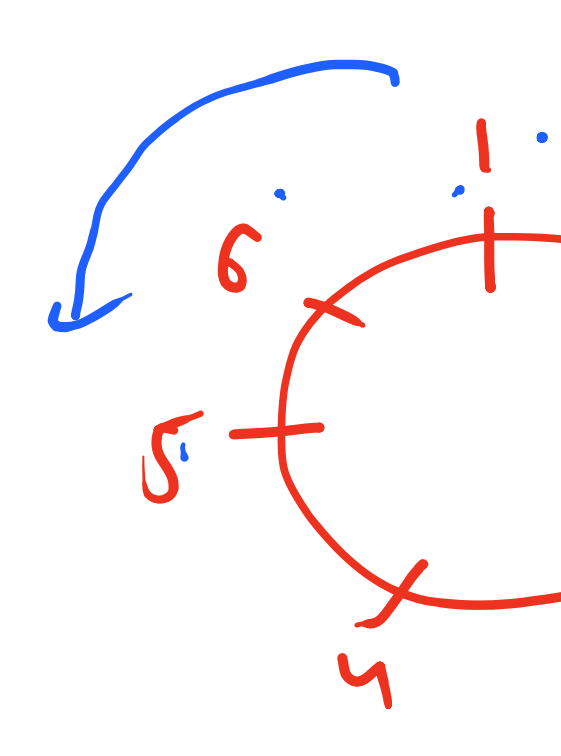
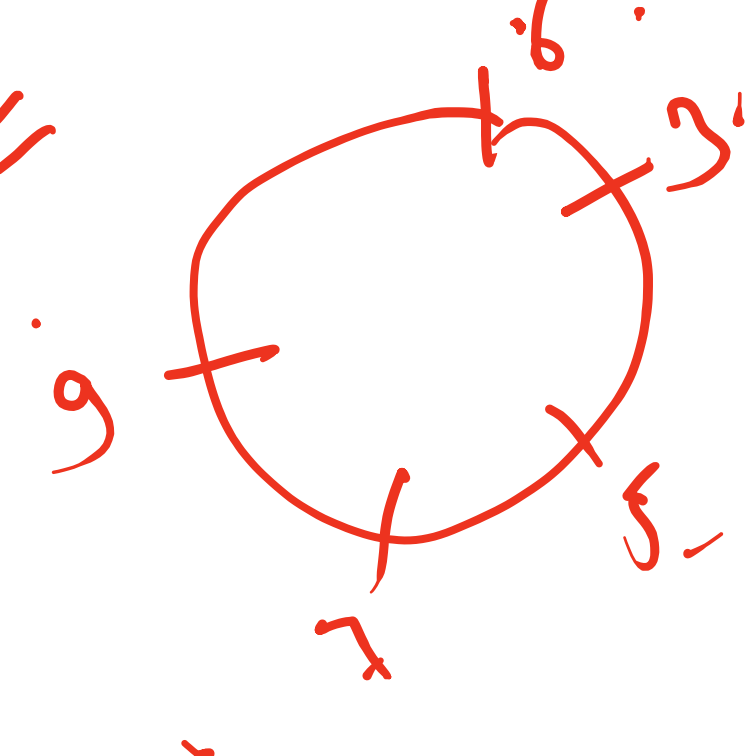
13 12 11 10 14



1 2 3 4 5 6

6 8 4 6 8 10

x=2



$j = i - x = 0 - 2 = -2$

$j = -2 = -1 + 4 = 3$

$j = 2 - 2 = 0$

$j = 3 - 2 = 1$

$j = 4 - 2 = 2$

$j = 5 - 2 = 3$